

Assignment 02 Report

Spatial Intelligence Through Graph Analysis

Course: Graph Machine Learning **Assignment:** Homework 02, Spatial Intelligence Part 1 **Date:** 6 June 2026

Abstract

This assignment investigates an architectural floor plan as a graph-based spatial system. A simplified OBJ outline is imported into TopologicPy, discretised with a regular grid, sliced into 250 topological cells, and converted into graphs representing spatial adjacency. Three analyses are then performed: shortest-path routing, closeness centrality, and betweenness centrality. The results identify the long internal corridor as the dominant circulation spine. Its central section is both the most globally integrated part of the plan and the part through which the greatest number of shortest routes pass. A grid-derived route of 24.84 units is reduced to 22.12 units after geometric straightening, a reduction of 2.72 units or approximately 10.95%. The study demonstrates how graph analysis can reveal spatial accessibility, hierarchy, and bottlenecks from architectural geometry.

1. Introduction

Architectural plans contain both geometric and relational information. Geometry describes the size and shape of spaces, while topology describes which spaces are connected and how movement can occur between them. Graph analysis provides a useful abstraction for studying the second type of information.

Navigable regions can be represented as vertices, adjacency as edges, and movement through the building as paths through the resulting graph.

The aim of this assignment is to construct a graph from a residential floor-plan outline and use it to answer three spatial questions:

1. What route connects two selected points through the accessible plan?

2. Which regions are most accessible from all other regions?
3. Which regions function as necessary passages between other regions?

These questions are addressed through shortest-path analysis, closeness centrality, and betweenness centrality.

2. Input Data and Tools

The source drawing is a residential plan organised around a long internal corridor, with repeated dwelling units on both sides and a lobby and vertical circulation zone toward the western end. The analysis uses `plan_5.obj`, a simplified accessible outline derived from the detailed drawing. Simplification is important because the analysis requires a clean, closed face rather than every line and annotation in the architectural drawing.

The implementation uses Python and TopologicPy. Key classes include `Topology` for importing and querying geometry, `Grid` for discretisation, `Graph` for topological conversion and network analysis, and `wire` for route geometry. The recorded notebook run used TopologicPy 0.9.20 and requires version 0.9.18 or newer.

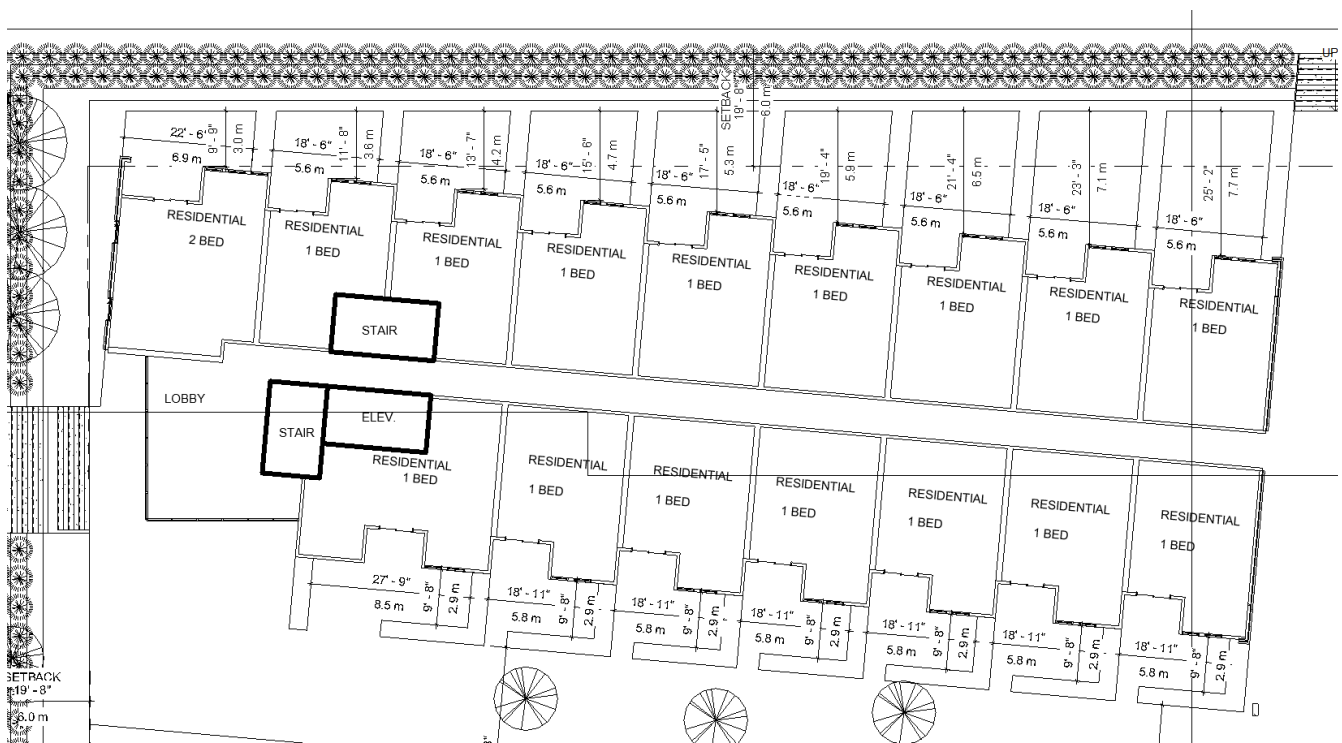


Figure 1. Source residential floor plan used to derive the simplified analysis boundary.

3. Methodology

3.1 Geometry Import and Validation

The OBJ file is loaded with `Topology.ByOBJPath`. The imported object is searched first for a face and then for a wire that can be converted into a face. This produces one valid floor face. Its bounding rectangle has the following recorded dimensions:

- x extent: 1.0 to 24.3 units
- y extent: 0.1 to 7.9 units
- width: 23.3 units
- length: 7.8 units

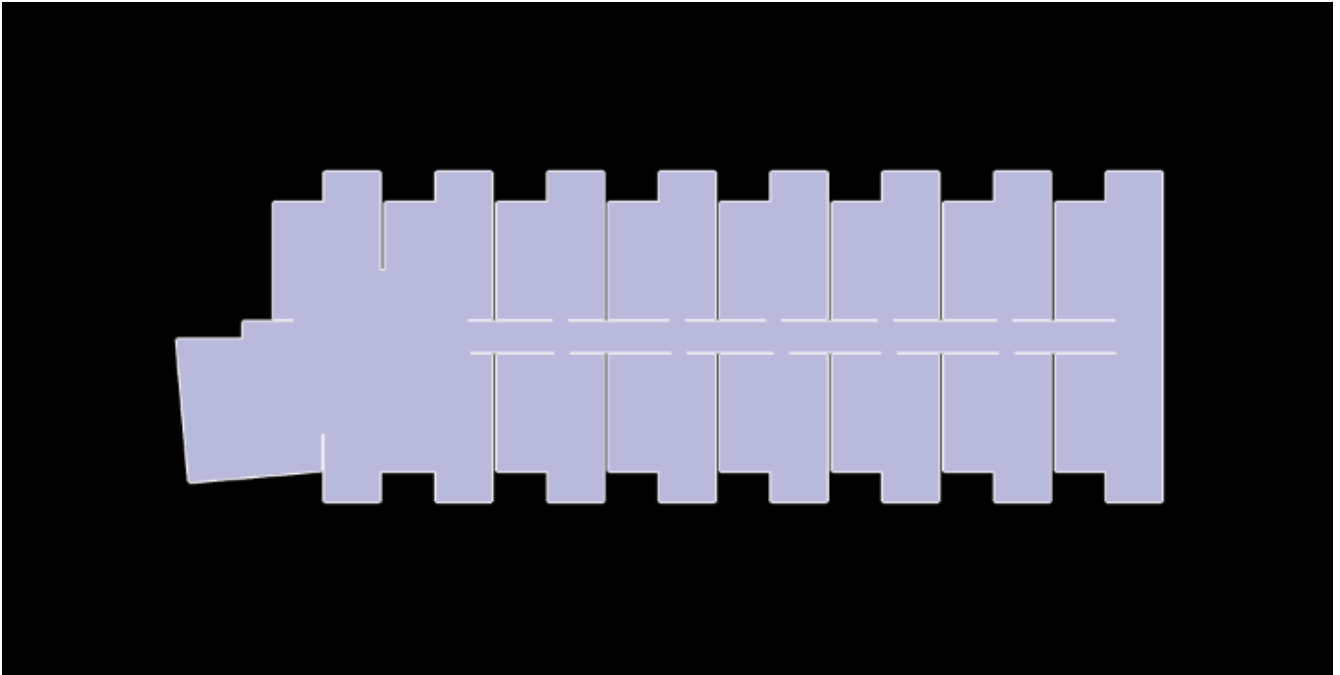


Figure 2. Simplified closed floor face imported from `pLan_5.obj`.

3.2 Grid Discretisation

A regular grid with a spacing of 1 model unit is clipped to the floor face. The floor is then sliced by the grid using `Topology.Slice`. Each resulting face receives a unique `face_id`, which is later used to transfer graph analysis values back to the corresponding spatial cell.

The operation generates 250 cells. This transforms continuous floor geometry into a finite set of comparable spatial units.

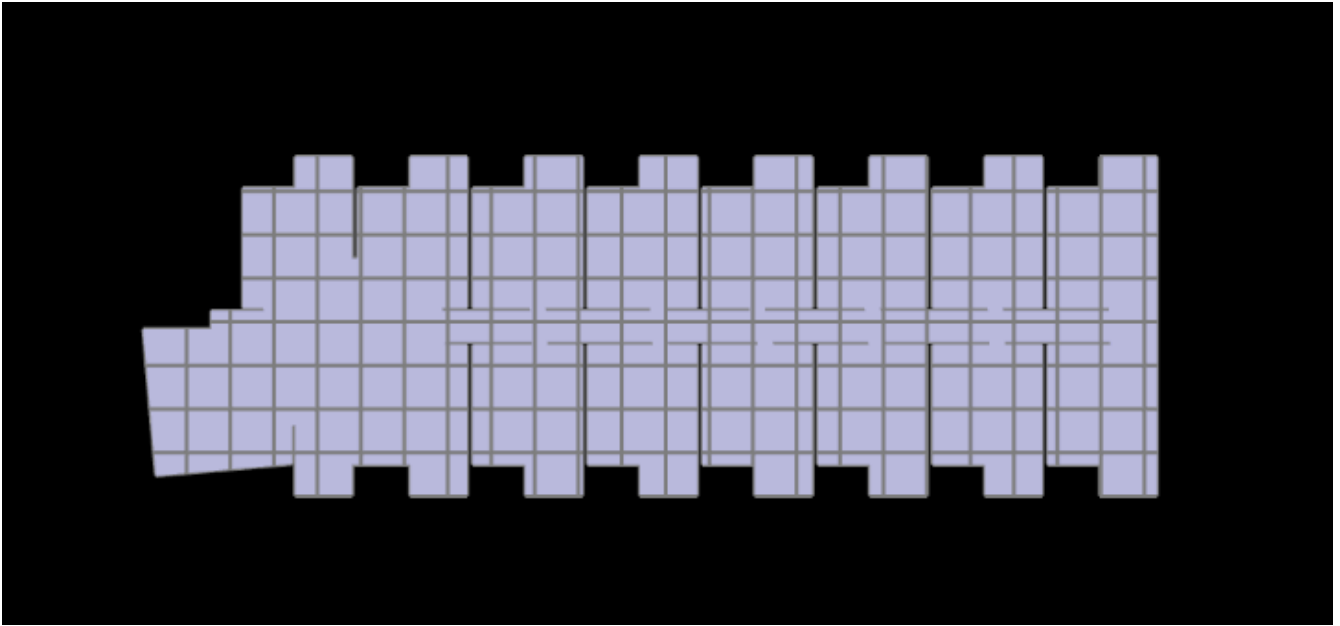


Figure 3. One-unit grid clipped to the accessible floor outline.

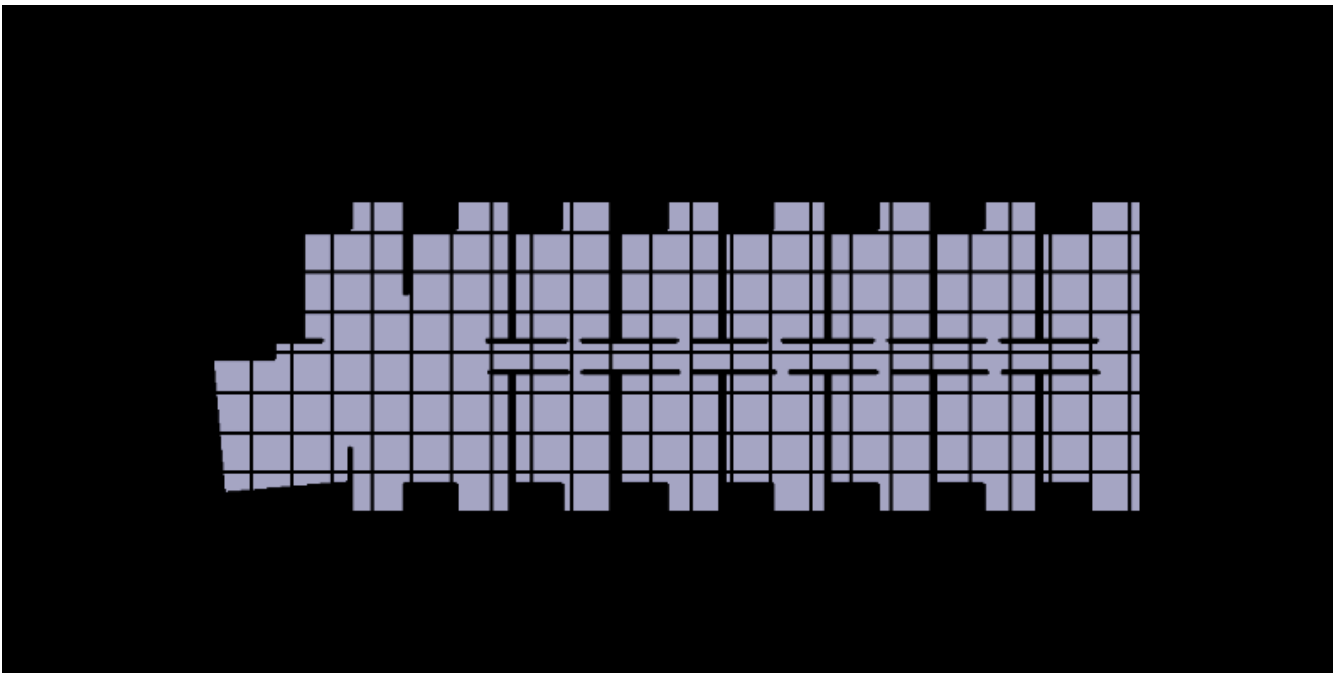


Figure 4. Topological shell produced by slicing the floor face with the grid.

3.3 Graph Construction

Two graphs are derived from the shell:

- A navigation graph is generated with shared-topology connections enabled.

It is used for route finding.

- An analysis graph is generated from the shell and used for centrality

calculations.

Graph vertices represent the discretised spatial cells, while graph edges represent traversable adjacency between cells. The graph therefore preserves the relational organisation of the plan while reducing its

geometric complexity.

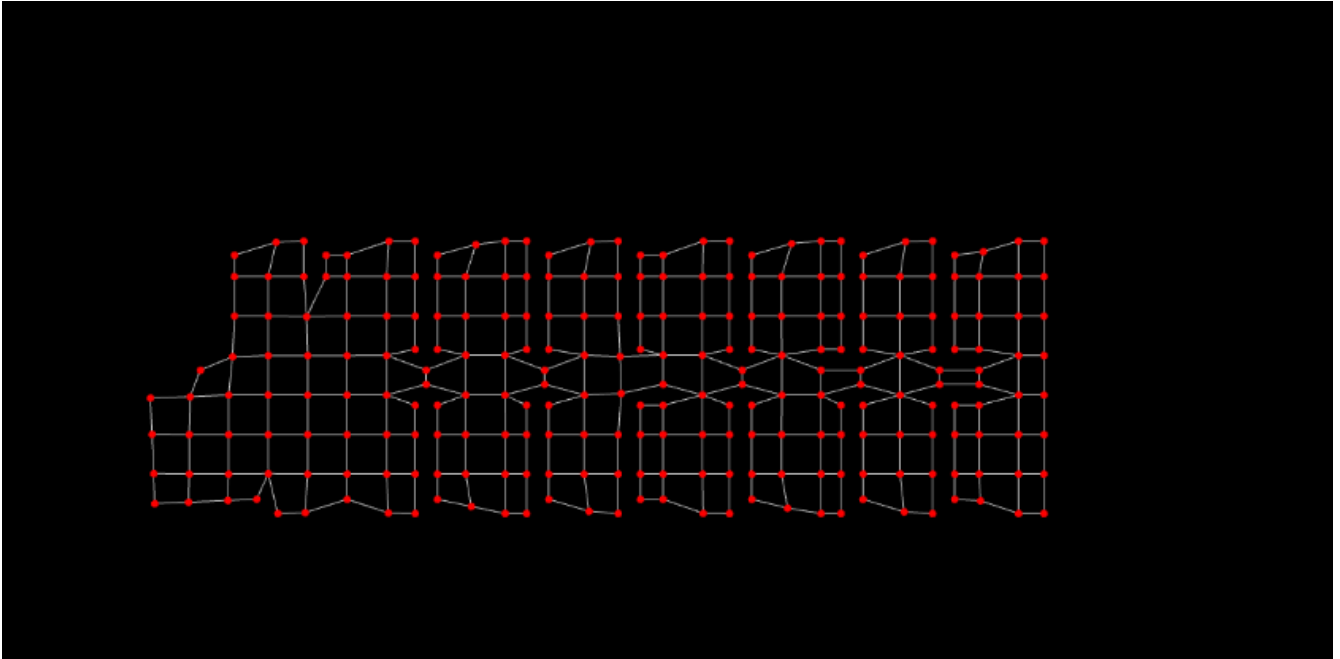


Figure 5. Analysis graph showing cell centres as vertices and adjacency as edges.

3.4 Shortest-Path Analysis

The start point is placed near the upper-western side of the plan and the end point near the lower-eastern side. A compiled routing graph is created and TopologicPy computes a shortest path between the endpoints. Because the graph follows discretised cells, the original route contains small zig-zag deviations. `Wire.Straighten` is then used to produce a shorter route that remains constrained by the floor face.

3.5 Centrality Analysis

Closeness centrality measures how near each graph vertex is to all other vertices through shortest-path distance. High closeness indicates strong global accessibility. In space syntax terminology, this is interpreted as integration.

Betweenness centrality measures the proportion of shortest paths between other vertex pairs that pass through a given vertex. High betweenness indicates that a location is important for through-movement. In space syntax this is related to choice.

Computed values are stored in the graph-vertex dictionaries and transferred back to cells using `face_id`, allowing the metrics to be visualised as continuous spatial heat maps.

4. Results

4.1 Quantitative Summary

Measure	Recorded result
Imported OBJ objects	1
Floor dimensions	23.3 x 7.8 units
Grid spacing	1 unit
Topological cells	250
Shortest-path computation time	0.15 s
Route-straightening time	0.96 s
Original grid path	24.84 units
Straightened path	22.12 units
Absolute reduction	2.72 units
Relative reduction	10.95%

The straightened route is approximately 10.95% shorter than the original grid-derived route. This difference quantifies part of the discretisation effect: movement constrained to graph transitions is less geometrically efficient than movement represented by a continuous line within the same boundary.

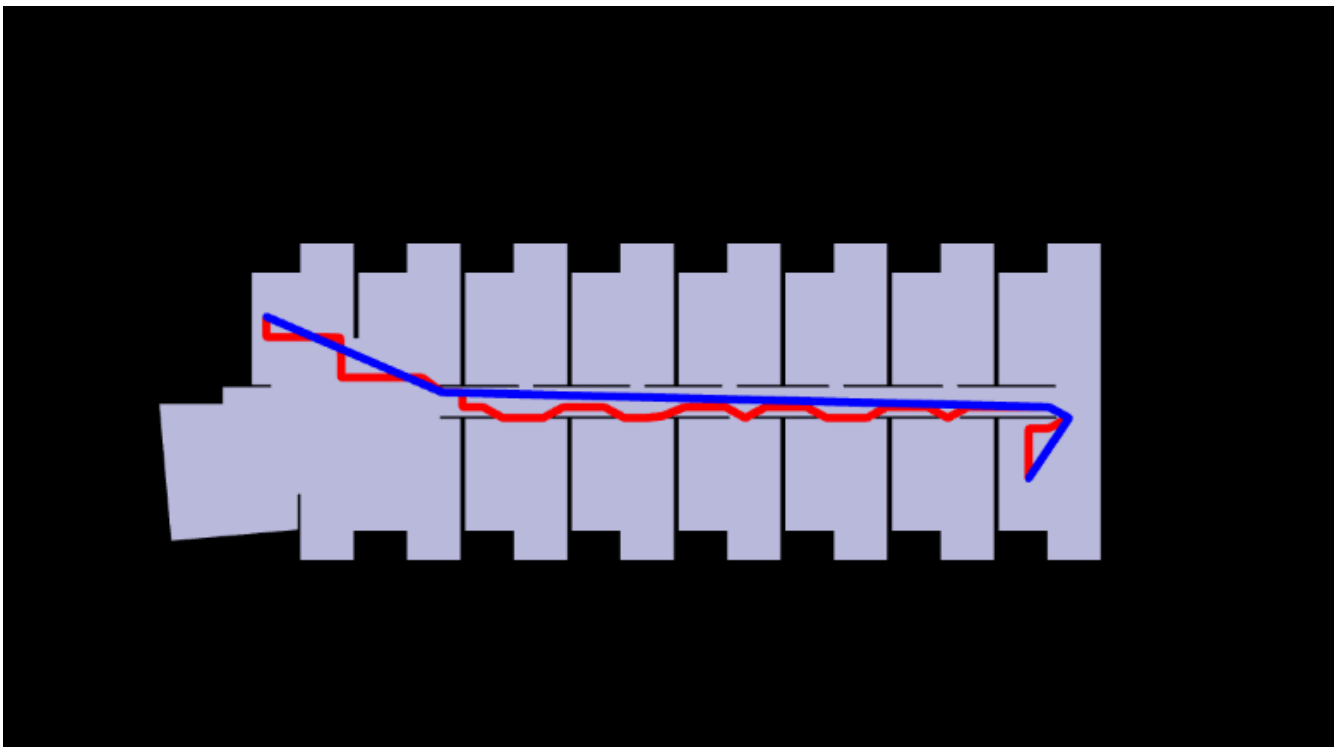


Figure 6. Grid-derived shortest path in red and straightened path in blue.

4.2 Closeness Centrality

The closeness map shows its highest values around the middle of the main corridor. Values decrease toward both ends of the building and into the deeper parts of individual dwelling bays. This pattern is consistent with the elongated geometry: central corridor cells minimise the total network distance to the largest number of other cells.

The result suggests that the central corridor is the most globally integrated location. From a design perspective, this area is well suited to shared wayfinding information or common functions because it can be reached with relatively few graph steps from the rest of the plan.

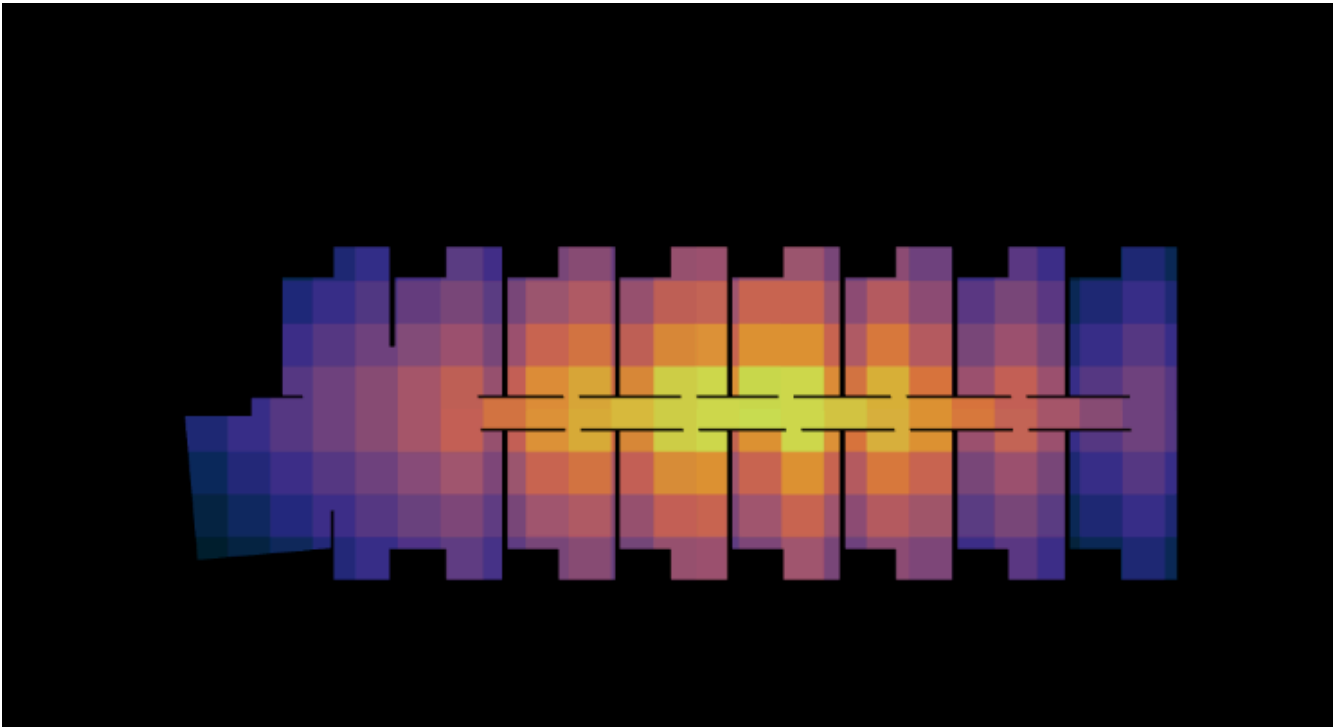


Figure 7. Closeness centrality heat map. Brighter central cells have stronger global integration.

4.3 Betweenness Centrality

The betweenness map concentrates high values along the main corridor, especially around its central segment. Peripheral cells have low values because they tend to be origins or destinations rather than necessary intermediate locations.

This result identifies a clear movement bottleneck. Routes between western and eastern areas, and between many opposing dwelling bays, must use the corridor. Consequently, disruption or congestion in this zone would affect a large proportion of potential journeys. The finding is relevant to circulation capacity, emergency egress, and the placement of obstacles or shared facilities.

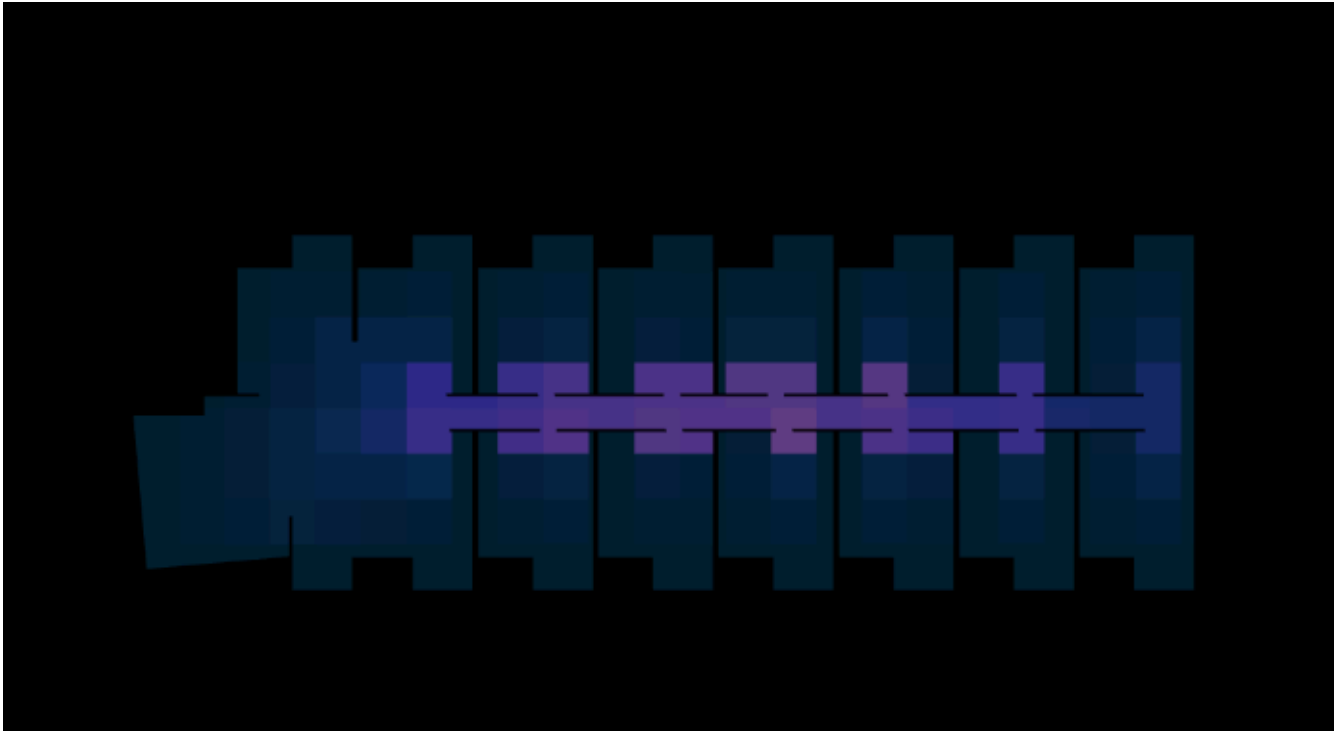


Figure 8. Betweenness centrality heat map. The corridor carries most cross-plan shortest routes.

5. Discussion

The three analyses describe complementary properties of the same spatial system. Shortest-path analysis evaluates one selected journey. Closeness centrality evaluates how efficiently each cell can reach the complete graph. Betweenness centrality evaluates how strongly each cell mediates journeys between all other pairs.

Together, the results reveal a strongly corridor-dominated topology. The repeated residential bays create local branches, while the horizontal corridor provides the only continuous east-west connection. This explains why central corridor cells score highly in both centrality measures. They are close to much of the graph and are also unavoidable for many cross-plan routes.

The western entrance and lobby area has lower global centrality than the middle of the corridor. Although it is functionally important as an entry, its position at one end of an elongated graph increases its average distance to eastern cells. This distinction shows that functional importance and topological centrality are related but not identical.

6. Limitations

First, the model depends on a 1-unit grid. A finer grid would represent boundaries and diagonal movement more accurately but would increase graph size and centrality computation time. A coarser grid would be faster but could remove narrow connections or exaggerate path error.

Second, the simplified floor face does not encode doors, room functions, occupancy, accessibility constraints, capacities, or direction-dependent costs. All graph connections are effectively treated as equally traversable. The analysis is therefore topological rather than a complete simulation of human movement.

Third, the route endpoints are manually chosen, so the path result represents one scenario rather than all origin-destination demands. Finally, the notebook visualises centrality comparatively but does not export a table of exact per-cell values, limiting detailed statistical comparison.

7. Future Work

The analysis could be extended by:

1. comparing several grid resolutions to test result stability;
2. modelling doors and restricted areas explicitly;
3. assigning weighted edges based on distance, width, or travel cost;
4. testing multiple origin-destination pairs and occupancy scenarios;
5. exporting cell-level metrics for statistical analysis; and
6. comparing graph predictions with observed pedestrian movement.

8. Conclusion

This assignment demonstrates a complete geometry-to-graph workflow for architectural spatial analysis. A simplified residential floor plan was discretised into 250 cells and converted into navigation and analysis graphs. The shortest-path experiment showed that geometric straightening reduced the grid route from 24.84 to 22.12 units. Closeness and betweenness centrality both identified the internal corridor, particularly its middle section, as the plan's principal spatial structure.

The analysis confirms that graph methods can expose accessibility, hierarchy, and bottlenecks that are not captured by geometric dimensions alone. Even with a simplified model, the results provide interpretable evidence about how the building supports movement and where its circulation network is most critical.

References

1. Hillier, B., and Hanson, J. (1984). *The Social Logic of Space*. Cambridge

University Press.

1. TopologicPy documentation. <https://topologicpy.readthedocs.io/>
2. NetworkX documentation, centrality algorithms.

<https://networkx.org/documentation/stable/reference/algorithms/centrality.html>